

What is Programming?



Just like we use Hindi or English to communicate with each other, we use programming languages like Python to communicate with computers.

Programming is the process of giving instructions to a computer so it can perform tasks, solve problems, and automate work.

What is Python?

Python is one of the easiest and most beginner friendly programming languages. Its syntax feels close to simple English which makes it easy to read, write, and understand.

Python is widely used in web development, automation, artificial intelligence, machine learning, data science, and software development.

Features of Python.

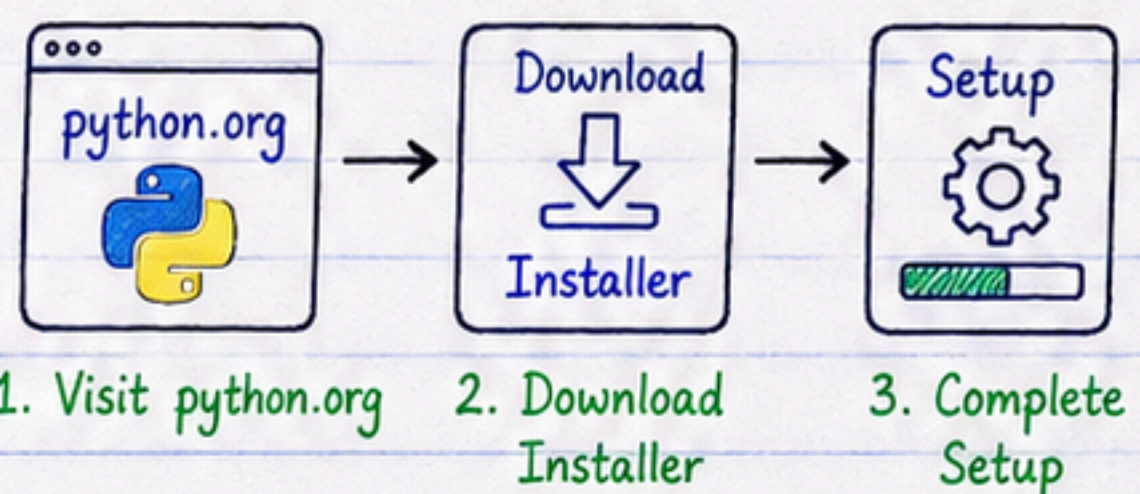
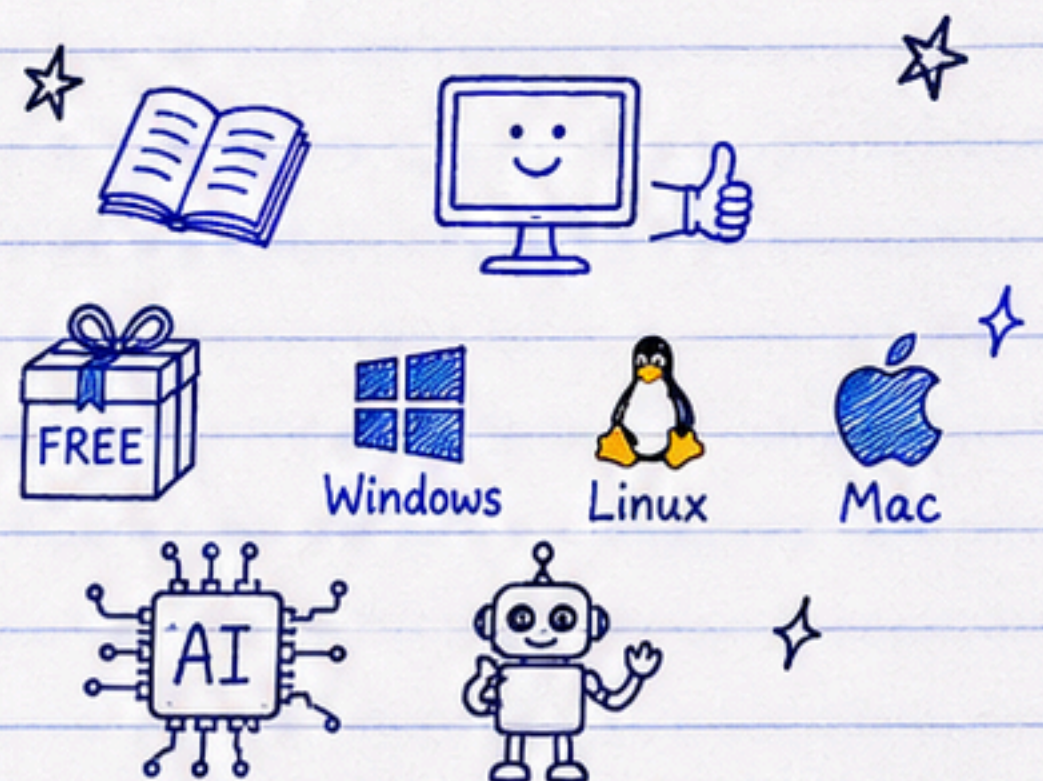
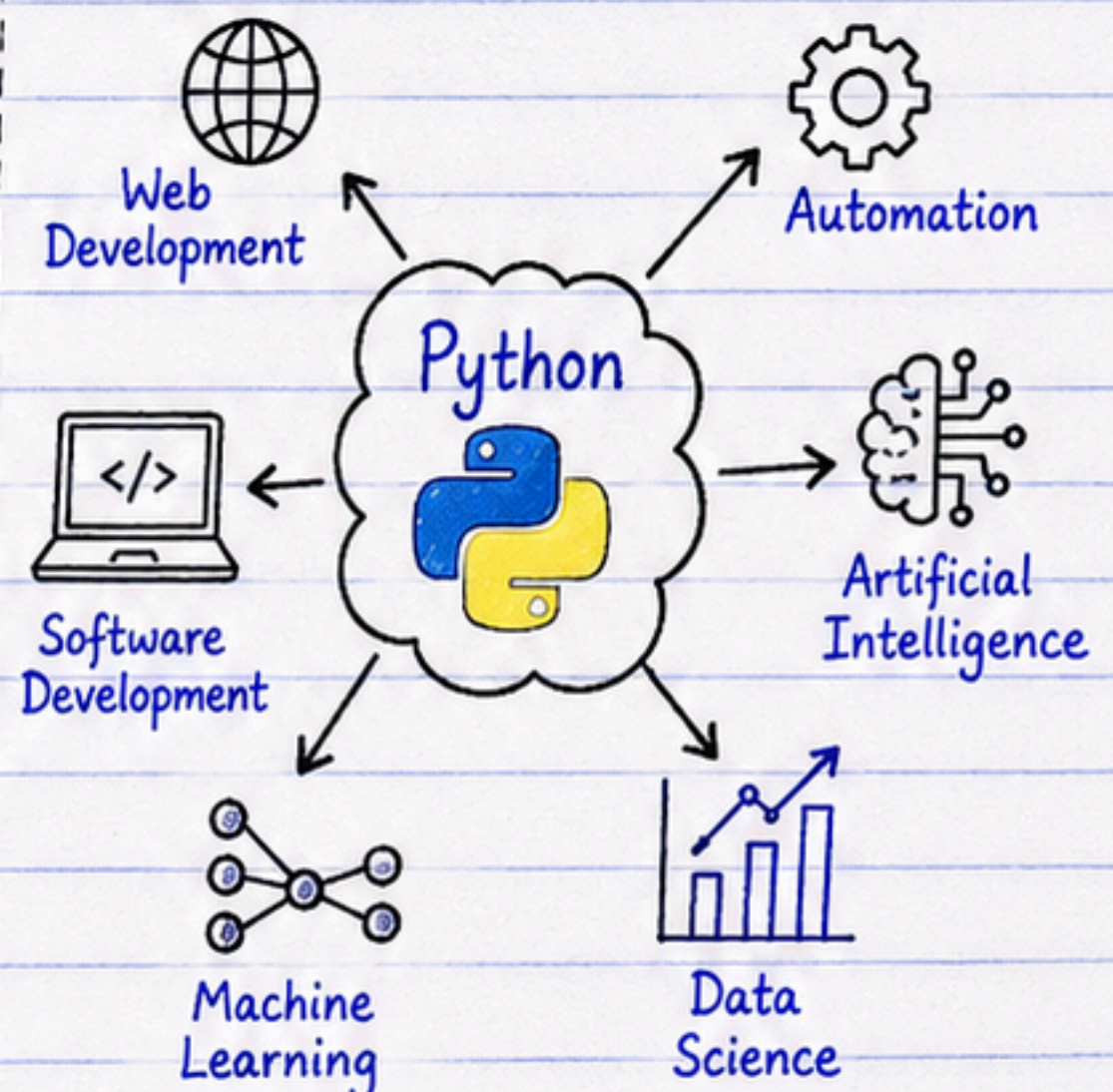
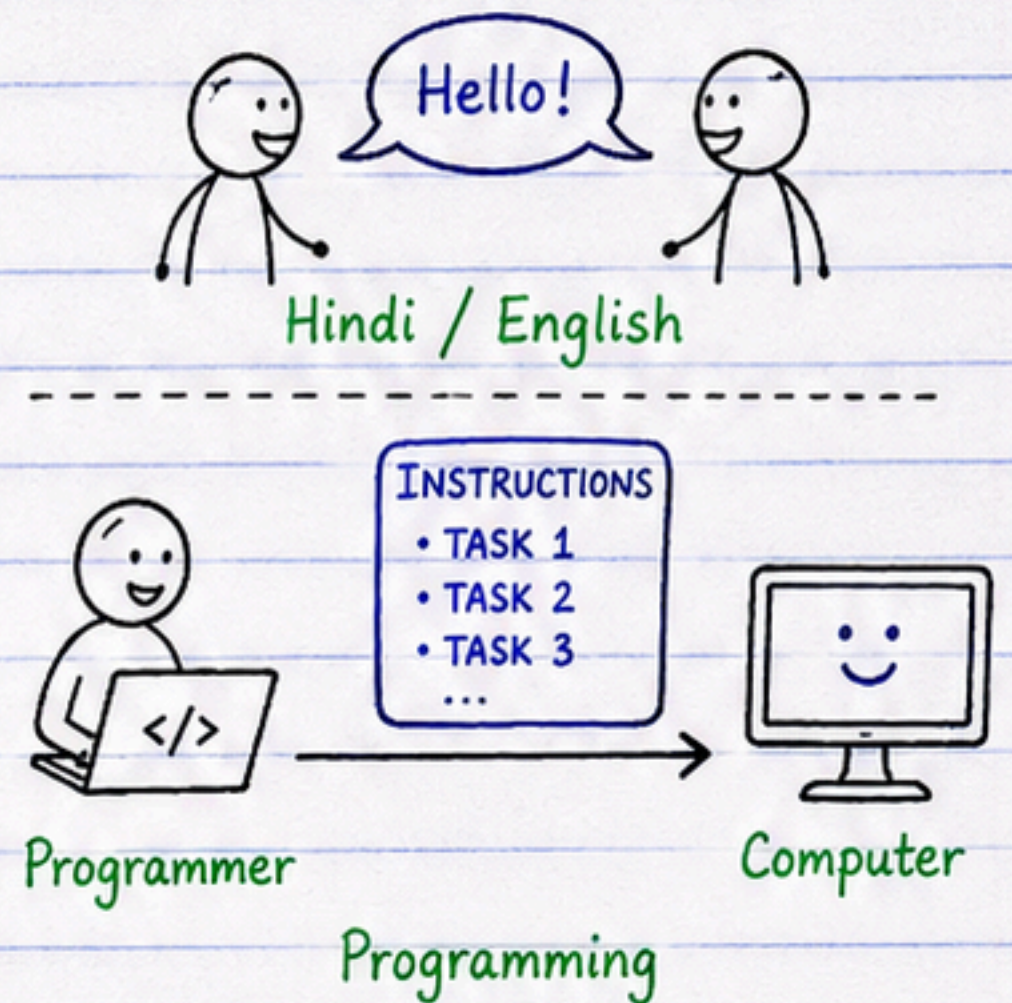
- 01 Easy to learn and beginner friendly syntax.
- 02 Free and open source programming language.
- 03 Works on Windows, Linux, and Mac systems.
- 04 Widely used in modern technologies and AI.

Installation.

GETTING STARTED

Python can be downloaded from python.org. Download the installer for your operating system and complete the setup process.

Just install it like a game

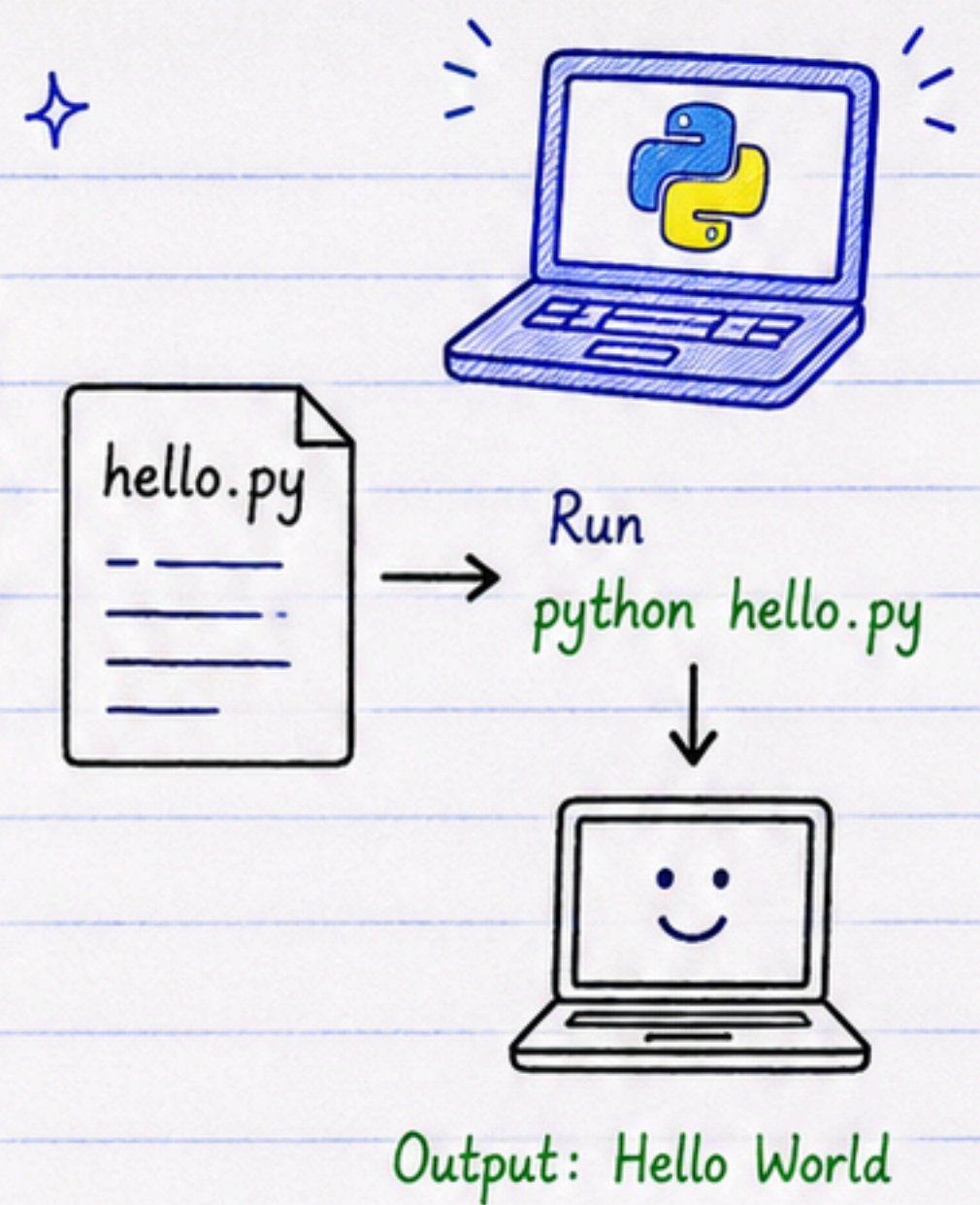


Modules, Comments & Pip

Let's write our very first python program.
Create a file called `hello.py` and paste the below code in it.

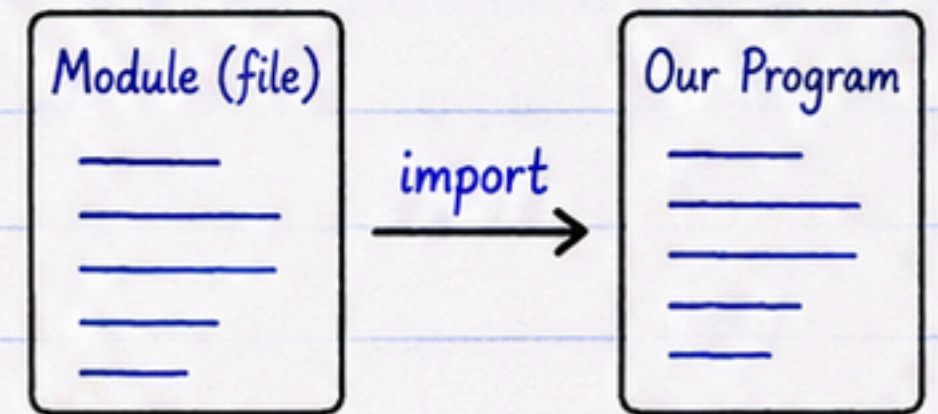
```
print("hello world") # print is a function
```

Execute this file by typing `python hello.py` and you will see `Hello World` printed on the screen.



★ Modules

A **module** is a file containing code written by somebody else which can be imported and used in our programs.



★ Pip

Pip is the **package manager** for Python. You can use pip to install external modules on your system.

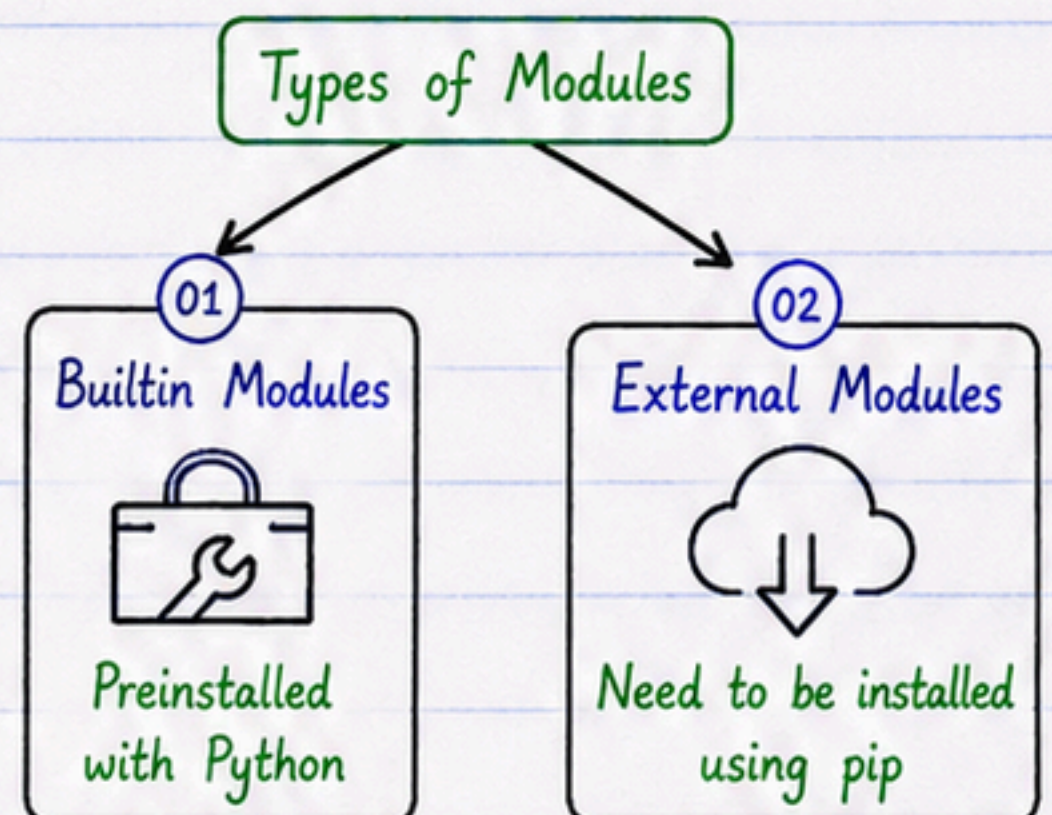
```
pip install flask
```



★ Types of Modules

There are **two types** of modules in Python.

- ① **Builtin Modules** are preinstalled with Python.
- ② **External Modules** need to be installed using **pip**.



Examples of builtin modules are `os` and `random`. Examples of external modules are `tensorflow` and `flask`.



Python as a Calculator

REPL MODE

Use Python directly from the terminal

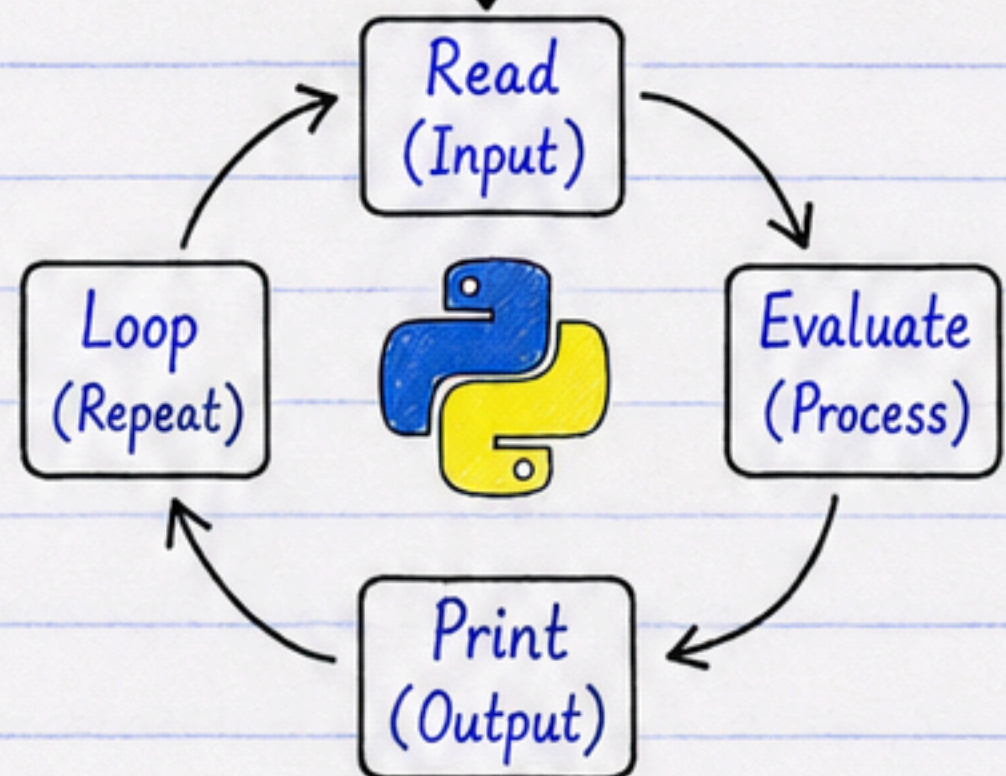
We can use Python as a calculator by typing `python` in the terminal.

This opens `REPL` which stands for `Read Evaluate Print Loop`.

```
$ python
>>> 2 + 3
5
>>> 10 / 2
5.0
>>> 7 * 4
28
>>> □
```

Wow!

REPL
(Read Evaluate Print Loop)



Comments

Comments are used to write something which the programmer **does not want to execute**.

They can be used to **explain code**, **mark author names**, **dates**, or **add notes**.

```
# Explain code
# Author: Alex
# Date: 20-05-2024
# This is a note
```

Types of Comments

There are **two types** of comments in Python.

① **Single Line Comments**

To write a single line comment, add a `#` symbol at the start of the line.

```
# This is a Single-Line Comment
```

Short and simple!

② **Multiline Comments**

To write multiline comments, you can use triple quotes.

```
"""
This is an amazing
example of a Multiline
comment!
"""
```

Useful for long notes!



Practice Set



1. Write a program to print **Twinkle twinkle little star** poem in python.

```
print("Twinkle twinkle little star,")
print("How I wonder what you are!")
print("Up above the world so high,")
print("Like a diamond in the sky.")
print("Twinkle twinkle little star,")
print("How I wonder what you are!")
```



2. Use REPL and print the **table of 5** using it.

REPL

```
$ python
>>> 5 * 1 = 5
>>> 5 * 2 = 10
>>> 5 * 3 = 15
>>> 5 * 4 = 20
>>> 5 * 5 = 25
>>> 5 * 6 = 30
>>> 5 * 7 = 35
>>> 5 * 8 = 40
>>> 5 * 9 = 45
>>> 5 * 10 = 50
>>> □
```



Python
as a
Calculator!

3. Install an **external module** and use it to perform an **operation** of your interest.

```
# Installing an external module
pip install math
# Using the module
import math
result = math.sqrt(25)
print("Square root of 25 is:", result)
```

External Module
math

Operation
Square Root
 $\sqrt{x}$

Output

5.0



4. Write a python program to print the **contents of a directory** using the **os** module. Search online for the function which does that.

```
import os

path = '.' # Current directory
contents = os.listdir(path)
for item in contents:
    print(item)
```



Lists all files
and folders in
a directory.

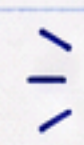
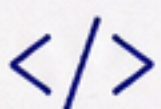
5. Label the program written in problem 4 with **comments**.

```
import os          # Import the os module

path = '.'         # Set the path to current directory
contents = os.listdir(path) # Get list of items in directory

# Iterate through each item and print it
for item in contents:
    print(item)    # Print the item (file or folder)
```

Comments
make code
easy to
understand!



Keep Practicing
Keep Coding!

